

Analysis of experimental errors in Mueller matrix channeled polarimeters

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This work analyzes the effect of experimental errors and measurement noise on Mueller matrix channeled spectropolarimeters. The main advantage of this type of polarimeter is the independence on temporal resolution, as it can be used as a snapshot polarimeter. The simulation of the polarimeters with experimental errors and two published extraction methods of the sample Mueller matrix are also presented. The Mueller matrix channeled spectropolarimeter (MMCS) setup consists of a mirrored Stokes channeled spectropolarimeter (SCS) as the polarization states generator (PSG) and a SCS as the polarization states analyzer (PSA). The SCS setup comprises two thick birefringent retarders followed by a horizontal linear polarizer. The effects of the thickness ratio of the retarders, the global retardance factor, retardance errors, axes alignment errors, and additive Gaussian noise are further studied to optimize the MMCS setups. In this work, we do not include a calibration procedure to improve the measured Mueller matrix parameters, but we study the sensitivity of the polarimeter to different configurations and error sources. © 2021 Optical Society of America

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1. INTRODUCTION

Mueller matrix polarimeters measure the full or partial Mueller matrix of a sample or an optical component. One of the difficulties of traditional polarimetric measurements is the requirement for quasi-monochromatic illumination, or a single measurement position, to be able to have a fast and simple measurement procedure. Channeled polarimeters (CP) with spatial or spectral channeling have been developed to overcome this type of limitation, and to permit snapshot polarimetry [1,2]. In particular, the spectroscopic analysis of the state of polarization (SOP) of light plays a major role in polarimetric and ellipsometric studies of dispersive materials [3,4]. Spectropolarimetry has been widely applied in various fields, such as remote sensing [5,6], material characterization [7,8], and synthesis of novel materials [9–11].

In this paper, we study the sensitivity of the measured Mueller matrix by a Mueller matrix channeled spectropolarimeter (MMCS) to possible configurations and experimental errors. Furthermore, we study the effects of the thickness ratio of the polarimeter retarders, the global retardance factor, retardance errors, axes alignment errors, and additive Gaussian noise, extending a previous study on the effect of these errors on a Stokes channeled spectropolarimeter (SCS) [12]. In this work, we do not include a calibration procedure to improve the measured Mueller matrix parameters, but we study the sensitivity of the polarimeter to the error sources. In Section 2, we present two published methods to extract the Mueller matrix of the sample.

In Section 3, we present the results of various simulations of different configurations and error sources. The paper ends with concluding remarks in Section 4.

2. CHANNELED SPECTROPOLARIMETRY

This paper analyzes Mueller matrix channeled polarimetry as a snapshot technique to measure spectral or spatial polarization characteristics [2]. Temporal misregistration, or intensity differences between time-sequential measurements not induced by polarization, can be a significant source of error in certain applications. Such misregistration can be caused by motion of the polarimeter or the sample [13,14].

CP techniques make use of polarization interference to amplitude modulate the Mueller matrix components onto either spectral or spatial carrier frequencies. CP enables the direct measurement of all 16 Mueller matrix components simultaneously, by performing the addition and subtraction optically [2].

Interference of different polarization components of the beam maintains the phase of each component within the complex amplitude, before the detector measures the intensity. Consequently, the amplitude and phase of the Mueller matrix elements are encoded within the amplitude and phase of the carrier frequency, enabling the magnitude and sign (or handedness) of the components to be extracted [14].

As CP is a snapshot method, there are no problems regarding image registration between the intensity measurements [14,15]. This avoids the creation of false polarization signatures that can occur in the spectrum or scene. Since a given Mueller parameter is calculated interferometrically and measured directly, image registration between several intensity measurements is unnecessary. Furthermore, since each parameter is modulated on coincident carrier frequencies, spatial or spectral registration between all the parameters is inherent. This significantly reduces the complexity of the Mueller matrix calculation over conventional polarimeters. However, these benefits come at a tradeoff, typically to the spatial and spectral resolution of the sensor [14]. This type of cosinusoidally modulated spectrum is widely used in the field of frequency-domain interferometry and is called a channeled spectrum [4].

The MMCS setup studied (see Fig. 1) is composed of a SCS as the PSA, whereas the PSG is a mirror image of the PSA. The SCS setup is composed of two thick birefringent retarders ($\underline{\mathbf{M}}_{R2}$), followed by a horizontal linear polarizer ($\underline{\mathbf{M}}_{P1}$) [16–18]. Then, the MMCS setup is given by

$$\underline{\mathbf{W}} = \underline{\mathbf{M}}_{P2}(0) \underline{\mathbf{M}}_{R4}(\phi_4, \pi/2) \underline{\mathbf{M}}_{R3}(\phi_3, 0) \underline{\mathbf{M}}_{R2}(\phi_2, 0) \underline{\mathbf{M}}_{R1}(\phi_1, \pi/2) \underline{\mathbf{M}}_{P1}(0), \quad (1)$$

where $\underline{\mathbf{M}}$ is the Mueller matrix of the sample. The retarders $\underline{\mathbf{M}}_{R2}$, $\underline{\mathbf{M}}_{R3}$ are aligned at 0° , the retarders $\underline{\mathbf{M}}_{R1}$, $\underline{\mathbf{M}}_{R4}$ at 45° , and the polarizers $\underline{\mathbf{M}}_{P1}$, $\underline{\mathbf{M}}_{P2}$ at 0° . The corresponding Mueller matrices are given by

$$\underline{\mathbf{M}}_{Pj}(0) = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad j = 1, 2, \quad (2)$$

$$\underline{\mathbf{M}}_{Rj}(\phi_j, 0) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \phi_j & \sin \phi_j \\ 0 & 0 & -\sin \phi_j & \cos \phi_j \end{pmatrix}, \quad j = 2, 3, \quad (3)$$

$$\underline{\mathbf{M}}_{Rj}(\phi_j, \pi/2) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \phi_j & 0 & -\sin \phi_j \\ 0 & 0 & 1 & 0 \\ 0 & \sin \phi_j & 0 & \cos \phi_j \end{pmatrix}, \quad j = 1, 4, \quad (4)$$

where ϕ_j is the retardance of the thick birefringent retarders given by $\phi_j = 2\pi\sigma\tau_j$ with $\tau_j = d_0d_jB$; σ , the wavenumber; τ_j , the optical path difference (OPD); d_0 , the global retardance factor in mm; d_j , the local retardance factor; and B , the birefringence of the retarder. The product d_0d_j is the retarder's thickness.

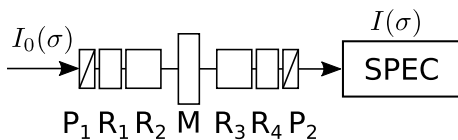


Fig. 1. Mueller matrix channeled spectropolarimeter setup comprising SCS as the PSA and a mirrored SCS as the PSG.

For this setup, an unpolarized light source with intensity $I_0(\sigma)$ and a polarization-insensitive detector are considered. The irradiance measured by the detector is given by

$$I(\sigma) = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix} \underline{\mathbf{W}} \begin{pmatrix} I_0 & 0 & 0 & 0 \end{pmatrix}^T. \quad (5)$$

Substituting the system matrix given by Eq. (1) and developing Eq. (5) yields

$$I(\sigma) = \frac{1}{4} I_0(\sigma) X(\sigma), \quad (6)$$

where

$$X(\sigma) = \begin{pmatrix} 1 & C\phi_4 & S\phi_3S\phi_4 & -C\phi_3S\phi_4 \end{pmatrix} \underline{\mathbf{M}} \begin{pmatrix} 1 \\ C\phi_1 \\ S\phi_1S\phi_2 \\ S\phi_1C\phi_2 \end{pmatrix}, \quad (7)$$

and $C\phi_j = \cos(\phi_j)$ and $S\phi_j = \sin(\phi_j)$. Therefore, the expected modulations of the MMCS are given by

$$\begin{aligned} X(\sigma) = & +m_{00} + m_{01}C\phi_1 \\ & + m_{02}S\phi_1S\phi_2 + m_{03}S\phi_1C\phi_2 \\ & + m_{10}C\phi_4 + m_{11}C\phi_1C\phi_4 \\ & + m_{12}S\phi_1S\phi_2C\phi_4 + m_{13}S\phi_1C\phi_2C\phi_4 \\ & + m_{20}S\phi_3S\phi_4 + m_{21}C\phi_1S\phi_3S\phi_4 \\ & + m_{22}S\phi_1S\phi_2S\phi_3S\phi_4 + m_{23}S\phi_1C\phi_2S\phi_3S\phi_4 \\ & - m_{30}C\phi_3S\phi_4 - m_{31}C\phi_1C\phi_3S\phi_4 \\ & - m_{32}S\phi_1S\phi_2C\phi_3S\phi_4 - m_{33}S\phi_1C\phi_2C\phi_3S\phi_4. \end{aligned} \quad (8)$$

It is observed that the MMCS works with up to 49 channels (including a central channel C_0 and 24 negative channels, see Table 1), and that the spectrometer resolution must be sufficient to resolve all these channels [19].

For the Mueller matrix components extraction, we studied two methods: (1) the Measurement matrix method [20], which we use only as a reference method and to characterize the polarimeters, and (2) the analytical channel splitting method proposed by Alenin and Tyo [1,2]. A channel splitting method, similar to one proposed by Oka and Kato for SCS systems [4], was developed by Hagen, Oka, and Dereniak [21] for a specific MMCS configuration, but is out of the scope of this paper.

A. Measurement Matrix Method

A series of analyzing polarization states (vectors) and generating vectors are predetermined

$$\underline{\mathbf{A}}_n = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \end{pmatrix}_n^T, \quad (9)$$

$$\underline{\mathbf{G}}_n = \begin{pmatrix} g_0 & g_1 & g_2 & g_3 \end{pmatrix}_n^T, \quad (10)$$

so the irradiance measured is given by

Table 1. MMCS Q-Matrix Template

Index	OPD	m_{00}	m_{01}	m_{02}	m_{03}	m_{10}	m_{11}	m_{12}	m_{13}	m_{20}	m_{21}	m_{22}	m_{23}	m_{30}	m_{31}	m_{32}	m_{33}	Factor
24	$+\tau_1 - \tau_2 - \tau_3 - \tau_4$											-1	$+j$			$-j$	-1	1/16
23	$+\tau_1 - \tau_2 - \tau_3 + \tau_4$											+1	$-j$			$+j$	+1	1/16
22	$+\tau_1 - \tau_2 + \tau_3 - \tau_4$											+1	$-j$			$-j$	-1	1/16
21	$+\tau_1 - \tau_2 + \tau_3 + \tau_4$											-1	$+j$			$+j$	+1	1/16
20	$+\tau_1 + \tau_2 - \tau_3 - \tau_4$											+1	$+j$			$+j$	-1	1/16
19	$+\tau_1 + \tau_2 - \tau_3 + \tau_4$											-1	$-j$			$-j$	+1	1/16
18	$+\tau_1 + \tau_2 + \tau_3 - \tau_4$											-1	$-j$			$+j$	-1	1/16
17	$+\tau_1 + \tau_2 + \tau_3 + \tau_4$											+1	$+j$			$-j$	+1	1/16
16	$+\tau_1 - \tau_3 - \tau_4$										-1				$-j$			1/8
15	$+\tau_1 - \tau_3 + \tau_4$										+1				$+j$			1/8
14	$+\tau_1 + \tau_3 - \tau_4$										+1				$-j$			1/8
13	$+\tau_1 + \tau_3 + \tau_4$										-1				$+j$			1/8
12	$+\tau_1 - \tau_2 - \tau_4$							+1	$-j$									1/8
11	$+\tau_1 - \tau_2 + \tau_4$							+1	$-j$									1/8
10	$+\tau_1 + \tau_2 - \tau_4$							-1	$-j$									1/8
9	$+\tau_1 + \tau_2 + \tau_4$							-1	$-j$									1/8
8	$+\tau_3 - \tau_4$									+1				$-j$				1/4
7	$+\tau_3 + \tau_4$									-1				$+j$				1/4
6	$+\tau_1 - \tau_4$						+1											1/4
5	$+\tau_1 + \tau_4$						+1											1/4
4	$+\tau_1 - \tau_2$			+1	$-j$													1/4
3	$+\tau_1 + \tau_2$			-1	$-j$													1/4
2	$+\tau_4$					+1												1/2
1	$+\tau_1$		+1															1/2
0	0	+1																1
-1	$-\tau_1$		+1															1/2
-2	$-\tau_4$					+1												1/2
-3	$-\tau_1 - \tau_2$			-1	$+j$													1/4
-4	$-\tau_1 + \tau_2$			+1	$+j$													1/4
-5	$-\tau_1 - \tau_4$						+1											1/4
-6	$-\tau_1 + \tau_4$						+1											1/4
-7	$-\tau_3 - \tau_4$									-1				$-j$				1/4
-8	$-\tau_3 + \tau_4$									+1				$+j$				1/4
-9	$-\tau_1 - \tau_2 - \tau_4$							-1	$+j$									1/8
-10	$-\tau_1 - \tau_2 + \tau_4$							-1	$+j$									1/8
-11	$-\tau_1 + \tau_2 - \tau_4$							+1	$+j$									1/8
-12	$-\tau_1 + \tau_2 + \tau_4$							+1	$+j$									1/8
-13	$-\tau_1 - \tau_3 - \tau_4$										-1				$-j$			1/8
-14	$-\tau_1 - \tau_3 + \tau_4$										+1				$+j$			1/8
-15	$-\tau_1 + \tau_3 - \tau_4$										+1				$-j$			1/8
-16	$-\tau_1 + \tau_3 + \tau_4$										-1				$+j$			1/8
-17	$-\tau_1 - \tau_2 - \tau_3 - \tau_4$											+1	$-j$			$+j$	+1	1/16
-18	$-\tau_1 - \tau_2 - \tau_3 + \tau_4$											-1	$+j$			$-j$	-1	1/16
-19	$-\tau_1 - \tau_2 + \tau_3 - \tau_4$											-1	$+j$			$+j$	+1	1/16
-20	$-\tau_1 - \tau_2 + \tau_3 + \tau_4$											+1	$-j$			$-j$	-1	1/16
-21	$-\tau_1 + \tau_2 - \tau_3 - \tau_4$											-1	$-j$			$-j$	+1	1/16
-22	$-\tau_1 + \tau_2 - \tau_3 + \tau_4$											+1	$+j$			$+j$	-1	1/16
-23	$-\tau_1 + \tau_2 + \tau_3 - \tau_4$											+1	$+j$			$-j$	+1	1/16
-24	$-\tau_1 + \tau_2 + \tau_3 + \tau_4$											-1	$-j$			$+j$	-1	1/16

$$I(\sigma_n) = \underline{\mathbf{A}}_n^T \underline{\mathbf{M}} \underline{\mathbf{G}}_n, \quad (11)$$

$$\underline{\mathbf{D}}'_n = \underline{\mathbf{A}}_n \otimes \underline{\mathbf{G}}_n \quad (13)$$

which can be shown is equivalent to

$$I(\sigma_n) = \underline{\mathbf{D}}'_n{}^T \underline{\mathbf{M}}', \quad (12)$$

where $\underline{\mathbf{D}}'_n$ is the dyad product $\underline{\mathbf{D}}'_n = \underline{\mathbf{A}}_n \underline{\mathbf{G}}_n^T$ reshaped into a vector

$$= (a_0 g_0 \dots a_0 g_3 \dots a_3 g_0 \dots a_3 g_3)_n^T, \quad (14)$$

and $\underline{\mathbf{M}}'$ is the Mueller matrix reshaped into a Mueller vector [2]

$$\underline{\mathbf{M}}' = (m_{00} m_{01} m_{02} m_{03} \dots m_{30} m_{31} m_{32} m_{33})^T. \quad (15)$$

Developing and rearranging Eq. (12) yields

$$\underline{\mathbf{I}} = \underline{\mathbf{W}}' \underline{\mathbf{M}}', \quad (16)$$

where

$$\underline{\mathbf{W}}' = (\underline{\mathbf{D}}'_0 \quad \underline{\mathbf{D}}'_1 \quad \dots \quad \underline{\mathbf{D}}'_{N-1})^T. \quad (17)$$

Therefore, $\underline{\mathbf{W}}'$ is a matrix containing the states of analysis-generation defined by the combination of the PSA and PSG of the MMCS [1,2].

The extraction of the Mueller matrix elements is achieved by inverting the process:

$$\underline{\mathbf{M}}' = \underline{\mathbf{W}}'^+ \underline{\mathbf{I}}, \quad (18)$$

where $\underline{\mathbf{W}}'^+$ is the pseudo-inverse of $\underline{\mathbf{W}}'$. However, this method considers the $\underline{\mathbf{M}}$ -matrix invariant with wavenumber; therefore, it is used only as a reference.

B. Analytical Channel Splitting Method

The inverse Fourier transform of $I(\sigma)$ [Eq. (6)] gives the autocorrelation function $C(\tau)$ [4], which describes the modulation channels in the τ -domain (OPD-space) of the spectrum of the Mueller matrix components we want to know. The Fourier transforms of the modulations present in Eq. (8) are

$$1(\sigma) \leftrightarrow \delta(\tau), \quad (19)$$

$$\cos(2\pi\sigma\tau_j) \leftrightarrow \frac{1}{2}[\delta(\tau + \tau_j) + \delta(\tau - \tau_j)], \quad (20)$$

$$\sin(2\pi\sigma\tau_j) \leftrightarrow \frac{j}{2}[\delta(\tau + \tau_j) - \delta(\tau - \tau_j)], \quad (21)$$

where δ , with no subscript, is the Dirac delta function, and τ is the Fourier transform variable of σ [1,2,14].

A construct is needed to recreate the modulations in the Fourier domain in an analytical form. This is obtained through a frequency phase matrix (FPM), which determines the functional form of the modulation and makes Fourier transforms a matter of looking up the correct row of a precalculated matrix [1,2]. Using the FPM, we obtain a map for each Mueller element splitting called the $\underline{\mathbf{Q}}$ -matrix [1], which maps the vectorized Mueller matrix into a channel vector $\underline{\mathbf{C}}$

$$\underline{\mathbf{C}} = \underline{\mathbf{Q}} \mathcal{F}^{-1}\{\underline{\mathbf{M}}'\}, \quad (22)$$

where $\underline{\mathbf{C}}$ is a matrix formed by concatenating the filtered channels of the autocorrelation function $C(\tau) = \mathcal{F}^{-1}\{I(\sigma)\}$ [2].

From the setup of interest (see Fig. 1), all θ 's are steps of 45° , which collapses $\cos(2\theta)$ and $\sin(2\theta)$ to either ± 1 or 0. As a result, only retardance contributes to the modulation [1]. We propose a template (see Table 1) for the $\underline{\mathbf{Q}}$ -matrix based on the local retardance factors. We construct the $\underline{\mathbf{Q}}$ -matrix by rearranging the rows of the template following the relative positions of the channels, given by comparison of the local retardance factors. The $\underline{\mathbf{Q}}$ -matrix has a cardinality $[N_C \times M]$, where

$M = 16$ and N_C is the number of channels or modulations, given by [1,2]

$$N_C = 1 + 2 \sum_{j=1}^4 d_j. \quad (23)$$

To extract the sample Mueller matrix an inverse map is required, which is a matrix that determines which channels have to be combined to extract the Mueller matrix components [1]. That inverse map corresponds to the pseudo-inverse $\underline{\mathbf{Q}}^+$,

$$\underline{\mathbf{M}}' = \underline{\mathbf{Q}}^+ \mathcal{F}\{\underline{\mathbf{C}}\}. \quad (24)$$

This method corresponds to the generalized channeled polarimetry methodology proposed by Alenin and Tyo [2].

It is important to note that a phase shift must be corrected when applying the Fourier transform to the filtered channels. For this, we analyze each possible channel and multiply it by a phase cancellation term corresponding to the retardance ϕ_j to each channel $C_j(\tau)$. This approach is mostly helpful for cases with crosstalk between channels (e.g., $\tau_1 - \tau_2 = \tau_2$).

3. SIMULATIONS, RESULTS, AND DISCUSSION OF MMCS CONFIGURATIONS

The general methodology for the MMCS simulations is as follows:

1. Select a sample (a Mueller matrix).
2. Define the MMCS configuration, including the spectral range $[\sigma_{\min}, \sigma_{\max}]$, the spectral resolution $\Delta\sigma$, the number of pixels N in the irradiance spectrum, the nominal global retardance factor, the nominal local retardance factors (thickness ratio), and the sources of error (retardance errors Δd_j , alignment errors ϵ_j , and Gaussian noise amplitude).
3. Run the methods reviewed to simulate the irradiance to be measured.
4. Run the inverse methods to extract the sample's Mueller matrix from the irradiance measured, for this purpose consider the irradiance obtained with Eq. (5) with no experimental errors.

The MMCS consists of a mirrored SCS (PSG) followed by the sample and a SCS (PSA), as discussed before (see Fig. 1). The transmission axis of the horizontal linear polarizer in the PSG is the system's reference (0°). For the spectrometer, a σ -wavenumber range from $1.4954 \times 10^4 \text{ cm}^{-1}$ to $1.8408 \times 10^4 \text{ cm}^{-1}$ (λ from 543 to 668 nm), with a sampling number N of 2048 pixels, and a corresponding wavenumber resolution of $(\sigma_{\max} - \sigma_{\min})/N$ were considered. In the simulation, the system was illuminated with an unpolarized light source with a varying spectral intensity. The high-order retarders were assumed to be made of quartz, and the birefringence was calculated using a model proposed by Ghosh [22]. The retardance shift for the retarders is between 84.68 and 106.10 rad/mm for the wavenumber range. The sample selected for the results reported is air (considered invariant with wavelength).

Reconstruction artifacts, associated with the fast Fourier transform (FFT) method applied, appear when extracting the Mueller matrix. This is a well-known effect, due to a lack

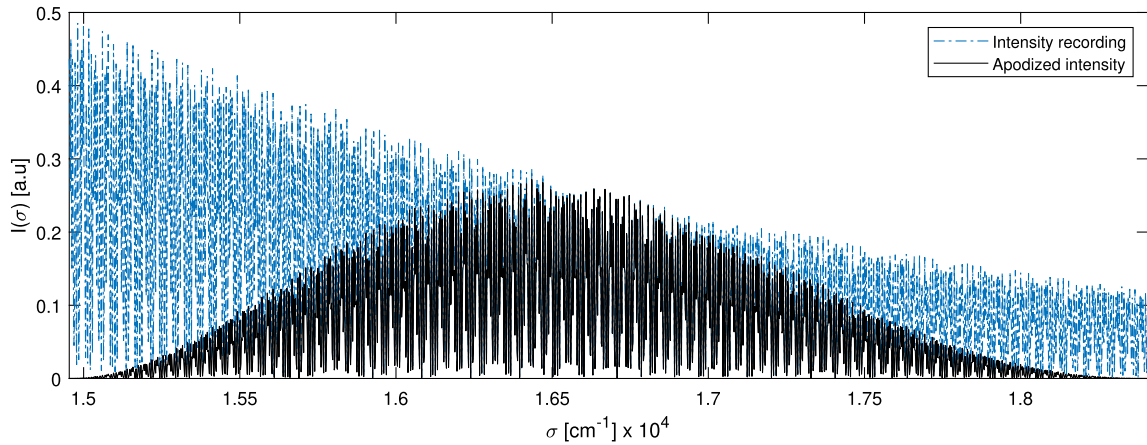


Fig. 2. Intensity recording (dashed light blue line) and apodized intensity (solid black line) using a configuration (1, 2, 5, 10), a global retardance factor $d_0 = 10$ mm, and an unpolarized light source with a varying spectral intensity, for the simulated air sample with SOP invariant with wavenumber.

of periodicity in the irradiance measured, contradicting the FFT supposition of a periodic function. Therefore, before computing the FFT, to prevent discontinuities in the periodic continuation, apodization has to be applied to the raw data (see Fig. 2) [11,23,24]. In this work, a Hann window was applied [25]:

$$\omega = \frac{1}{2} \left(1 - \cos \frac{2\pi n}{N} \right), \quad n = 0, 1, 2, \dots, N-1. \quad (25)$$

However this step may aggravate the discrepancies at the edges of the spectrum, where the division is nearly zero [23].

To evaluate the performance of the MMCS setups of interest, three figures of merit were considered: (1) the root mean square error (RMS) of the extracted elements m_{ij} against their corresponding nominal values $m_{ij,nom}$, given by

$$\text{RMS}(m_{ij}) = \left[N^{-1} \sum_{n=1}^N (m_{ij} - m_{ij,nom})^2 \right]^{1/2}, \quad (26)$$

(2) the condition number (CN) of the $\underline{\underline{W}}$ -Mueller matrix and the $\underline{\underline{Q}}$ -matrix of the nominal setup, and (3) the equally weighted variance (EWV) [26] for the same matrices given by

$$\text{EWV}(\underline{\underline{A}}) = \text{Tr}[(\underline{\underline{A}}^+)^T \underline{\underline{A}}^+], \quad (27)$$

where $\underline{\underline{A}}^+$ is the pseudo-inverse of $\underline{\underline{A}}$. These metrics are used to evaluate the noise immunity of the MMCS setups [27]. For this work, an extracted parameter was considered to be immune when RMS values were < 0.01 . To avoid the discrepancies at the edges of the spectra, a consequence of apodization [23], the RMS was calculated for a reduced wavenumber range from $1.5 \times 10^4 \text{ cm}^{-1}$ to $1.83 \times 10^4 \text{ cm}^{-1}$.

A. Thickness Ratio

Six retarder thickness ratios were considered: (1, 2, 3, 5), (1, 2, 4, 8), (1, 2, 5, 10), (1, 4, 2, 9), (2, 1, 4, 11), and (2, 1, 5, 12), with a global retardance factor $d_0 = 10$ mm. Figure 3 shows the corresponding autocorrelation function $|C(\tau)|$ for these thickness ratios. Figure 4 shows the EWV and CN plots for the

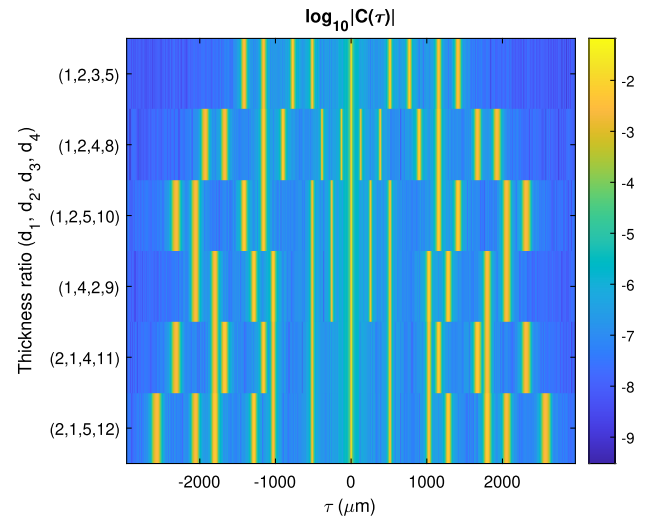


Fig. 3. Autocorrelation function $|C(\tau)|$ for the MMCS with different thickness ratios and a global retardance factor $d_0 = 10$ mm.

thickness' ratios of interest. The normalized Mueller matrix elements (m_{ij}/m_{00}) for air are shown in Fig. 5, and the RMS

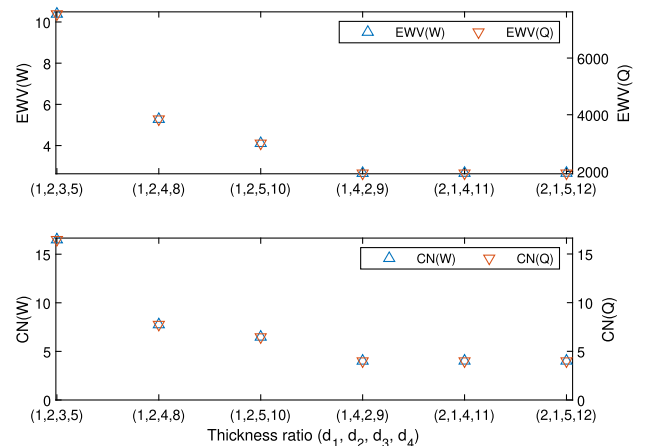


Fig. 4. EWV and CN of the $\underline{\underline{W}}$ - and $\underline{\underline{Q}}$ -matrix for the MMCS configurations (1, 2, 3, 5), (1, 2, 4, 8), (1, 2, 5, 10), (1, 4, 2, 9), (2, 1, 4, 11), and (2, 1, 5, 12) and a global retardance factor $d_0 = 10$ mm.

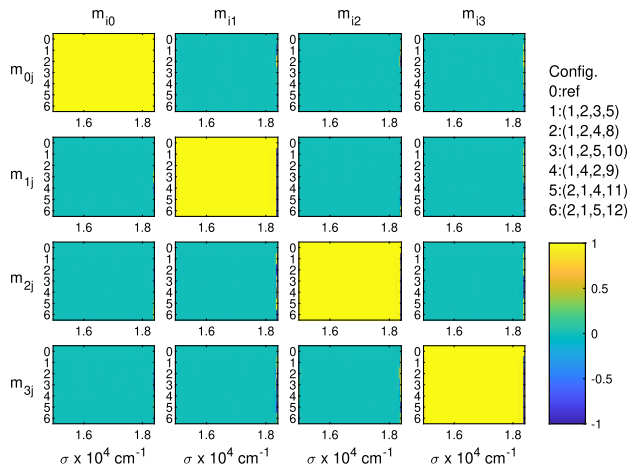


Fig. 5. Normalized Mueller matrix elements (m_{ij}/m_{00}) for the configurations (1, 2, 3, 5), (1, 2, 4, 8), (1, 2, 5, 10), (1, 4, 2, 9), (2, 1, 4, 11), and (2, 1, 5, 12) with a global retardance factor $d_0 = 10$ mm.

values for the extracted Mueller matrix $\underline{\mathbf{M}}$ and its elements m_{ij} are shown in Table 2. The configurations (1, 4, 2, 9), (2, 1, 4, 11), and (2, 1, 5, 12) have the lowest EWV and CN values, and the last two along with the configuration (1, 2, 5, 10) have the smallest total RMS values.

B. Global Retardance Factor

The MMCS configuration (1, 2, 5, 10) was used from this test forward, as it has been studied before [1,21]. Inspired by the dual-rotating-retarder Mueller matrix polarimeter [28], Hagen *et al.* chose a 5:1 ratio of thickness for the pair of analyzing retarders to the generating pair to give a compact result. Furthermore, Hagen *et al.* selected a 2:1 ratio of thicknesses for the retarders within each pair, giving a set of thicknesses designated as a (1, 2, 5, 10) configuration [21].

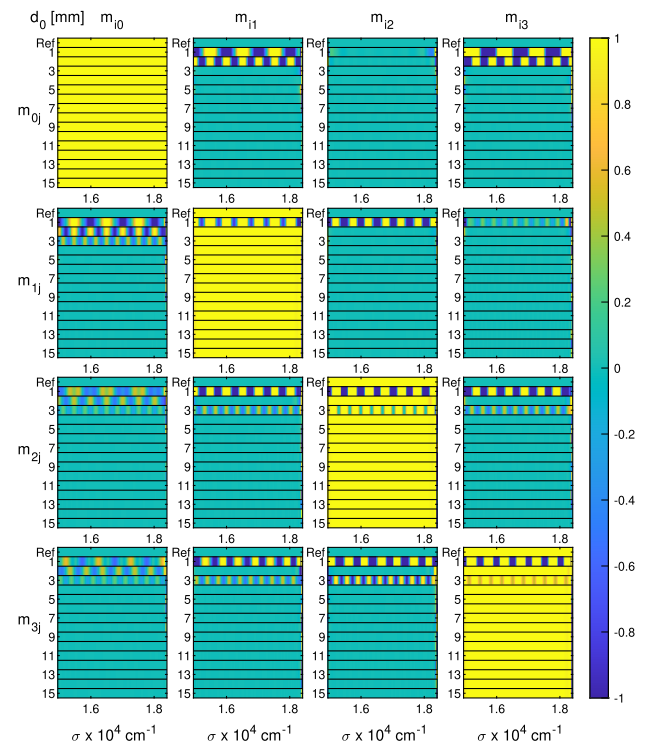


Fig. 6. Normalized Mueller matrix elements (m_{ij}/m_{00}) for d_0 from 1 to 15 mm, with a configuration (1, 2, 5, 10).

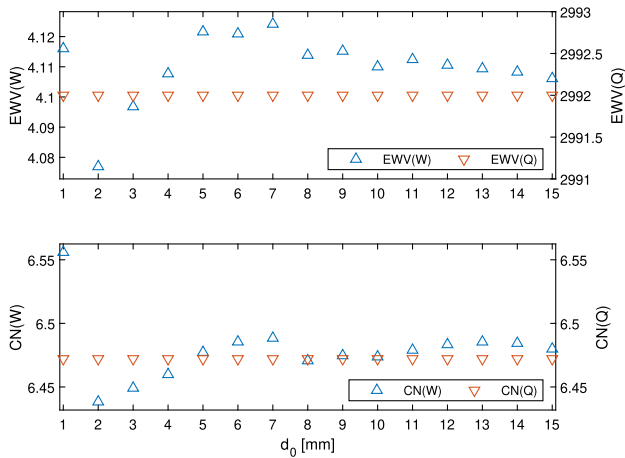
The extracted Mueller matrix using a global retardance factor d_0 from 1 to 15 mm is shown in Fig. 6. From the RMS values (see Table 3), it is observed that the MMCS performance generally improves when the global retardance factor is increased. Furthermore, the lowest total RMS is obtained with $d_0 = 15$ mm, and a total RMS error < 0.01 is achieved for $d_0 \geq 4$ mm. Considering this condition, the lowest EWV ($\underline{\mathbf{W}}$) is obtained with $d_0 = 15$ mm and the lowest CN ($\underline{\mathbf{W}}$), with $d_0 = 4$ mm (see Fig. 7).

Table 2. RMS of Extracted Mueller Matrix $\underline{\mathbf{M}}$ and Its Elements m_{ij} (Air Sample) Using Different Thickness Ratio Configurations (d_1, d_2, d_3, d_4)

	(1, 2, 3, 5)	(1, 2, 4, 8)	(1, 2, 5, 10)	(1, 4, 2, 9)	(2, 1, 4, 11)	(2, 1, 5, 12)
$\underline{\mathbf{M}}$	3.63 E-03	4.46 E-03	3.03 E-03	3.05 E-03	3.02 E-03	2.96 E-03
m_{00}	7.45 E-03	7.47 E-03	7.45 E-03	7.45 E-03	7.45 E-03	7.45 E-03
m_{01}	1.45 E-03	2.30 E-03	2.01 E-04	1.22 E-04	4.32 E-05	5.38 E-05
m_{02}	3.65 E-04	4.42 E-03	2.10 E-04	1.62 E-04	1.18 E-04	3.32 E-04
m_{03}	1.38 E-03	4.84 E-03	1.98 E-04	2.05 E-04	4.37 E-04	4.28 E-04
m_{10}	1.56 E-04	2.87 E-04	4.49 E-04	2.53 E-04	3.05 E-05	2.70 E-05
m_{11}	1.23 E-03	3.09 E-03	1.22 E-03	1.29 E-03	4.73 E-03	3.05 E-03
m_{12}	9.37 E-05	1.16 E-04	3.66 E-05	9.13 E-04	1.85 E-03	2.27 E-04
m_{13}	3.48 E-03	3.65 E-03	3.34 E-03	2.90 E-03	3.59 E-03	3.64 E-04
m_{20}	8.39 E-05	3.48 E-04	3.30 E-04	1.14 E-04	1.17 E-04	2.89 E-04
m_{21}	5.76 E-03	5.08 E-03	2.32 E-03	3.61 E-03	1.99 E-03	2.60 E-04
m_{22}	8.65 E-03	7.85 E-03	3.24 E-03	5.34 E-03	2.95 E-03	1.76 E-03
m_{23}	2.54 E-03	1.95 E-03	2.23 E-03	3.91 E-03	2.97 E-03	6.33 E-03
m_{30}	5.17 E-05	8.45 E-05	1.35 E-04	3.62 E-04	1.02 E-04	4.28 E-04
m_{31}	2.57 E-03	4.82 E-03	4.48 E-03	3.12 E-03	3.44 E-03	2.61 E-04
m_{32}	2.50 E-03	8.42 E-03	5.92 E-03	3.08 E-03	3.02 E-03	4.35 E-03
m_{33}	3.15 E-03	2.50 E-03	1.54 E-03	2.49 E-03	2.99 E-03	3.53 E-03

Table 3. RMS $\times 10^2$ of Extracted Mueller Matrix $\mathbf{M}$ and its Elements m_j (Air Sample) for the Global Retardance Factor d_0 from 1 to 15 mmGlobal retardance factor d_0 mm

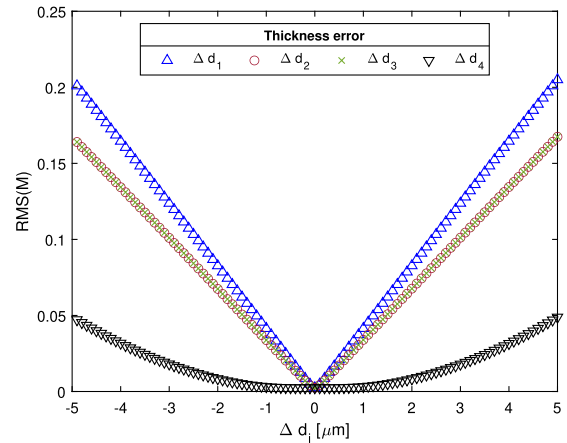
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\mathbf{M}$	164.87	61.63	32.45	0.52	0.61	0.44	0.37	0.35	0.31	0.31	0.25	0.28	0.30	0.28	0.24
m_{00}	17.78	0.84	0.69	0.73	0.74	0.74	0.74	0.74	0.74	0.75	0.74	0.75	0.75	0.75	0.75
m_{01}	121.31	124.53	2.87	0.69	1.61	0.46	0.23	0.28	0.18	0.05	0.10	0.03	0.02	0.04	0.02
m_{02}	11.88	2.26	1.06	0.34	0.25	0.28	0.03	0.17	0.02	0.07	0.02	0.02	0.02	0.02	0.01
m_{03}	200.97	193.65	0.62	1.36	1.15	0.27	0.39	0.11	0.06	0.04	0.12	0.02	0.02	0.02	0.01
m_{10}	69.45	70.91	35.41	0.14	0.54	0.52	0.14	0.08	0.06	0.07	0.04	0.02	0.04	0.01	0.01
m_{11}	148.66	1.66	0.12	0.15	0.18	0.83	0.23	0.12	0.22	0.40	0.23	0.39	0.12	0.17	0.32
m_{12}	160.35	1.60	0.30	0.14	0.12	0.02	0.03	0.01	0.01	0.02	0.01	0.01	0.00	0.00	0.01
m_{13}	23.93	1.89	0.29	0.41	0.44	0.15	0.37	0.31	0.17	0.18	0.07	0.23	0.33	0.30	0.20
m_{20}	39.39	35.65	17.69	0.07	0.15	0.05	0.12	0.09	0.04	0.08	0.05	0.03	0.03	0.01	0.01
m_{21}	176.65	4.75	35.33	0.24	0.34	0.13	0.29	0.38	0.18	0.27	0.20	0.30	0.23	0.36	0.17
m_{22}	360.05	9.93	70.68	0.35	0.61	0.26	0.51	0.56	0.58	0.54	0.18	0.36	0.32	0.52	0.22
m_{23}	179.68	3.04	35.60	0.43	0.25	0.22	0.22	0.13	0.31	0.30	0.27	0.14	0.22	0.19	0.20
m_{30}	40.73	35.27	17.70	0.10	0.17	0.11	0.19	0.08	0.05	0.02	0.04	0.04	0.01	0.01	0.01
m_{31}	114.11	3.88	35.27	0.24	0.15	0.26	0.45	0.39	0.26	0.21	0.07	0.30	0.45	0.20	0.09
m_{32}	229.42	7.44	70.41	0.83	0.31	0.94	0.68	0.73	0.43	0.45	0.23	0.34	0.59	0.18	0.24
m_{33}	253.23	1.88	35.34	0.29	0.43	0.41	0.39	0.20	0.41	0.32	0.37	0.17	0.15	0.23	0.26

**Fig. 7.** EWW and CN of the W - and Q -matrix of a MMCS with a (1, 2, 5, 10)-configuration against the global retardance factor d_0 from 1 to 15 mm.

C. Retardance Error

Because of the manufacturing tolerance, the thickness of high-order retarders and their retardance may deviate from design values [11]. Therefore, a fabrication tolerance Δd_j of $\pm 5 \mu\text{m}$ was considered, and it was assumed that the plates have completely flat and parallel faces. The fabrication tolerance was taken from manufacturers tolerances offered for custom thick birefringent quartz plates.

A global retardance factor of $d_0 = 15 \text{ mm}$ was considered from this test forward, because the lowest total RMS error was obtained with this value (see Table 3). Figure 8 shows the total RMS plots for the extracted Mueller matrix, considering the thickness error from the four retarders. It is observed that the extraction process is sensitive to the thickness error of the first three retarders, and it is about four times less sensitive to the thickness error of the fourth retarder. Immunity is achieved

**Fig. 8.** Total RMS ($\mathbf{M}$) error against the thickness error Δd_j of the j -th retarder, for $j = 1, 2, 3, 4$.

within a fabrication tolerance of up to $\pm 2 \mu\text{m}$ for the fourth retarder. This tolerance is much smaller for the first three retarders.

D. Alignment Error

The assembly process of an instrument is not perfect; therefore, the alignment errors (ϵ_j) of high-order retarders are unavoidable [11]. An alignment tolerance of up to $\pm 5^\circ$ was considered for the retarders and the second polarizer.

Figure 9 shows the total RMS plots for the extracted Mueller matrix, considering the alignment error of the four retarders ($\epsilon_1 - \epsilon_4$) and the second polarizer (ϵ_5). The extraction is acceptable within the tolerance of $\pm 5^\circ$. It is observed that the extraction process is more sensitive to the alignment error on the first retarder, for this retarder immunity is achieved between $\pm 0.6^\circ$. The performance for the other three retarders and the

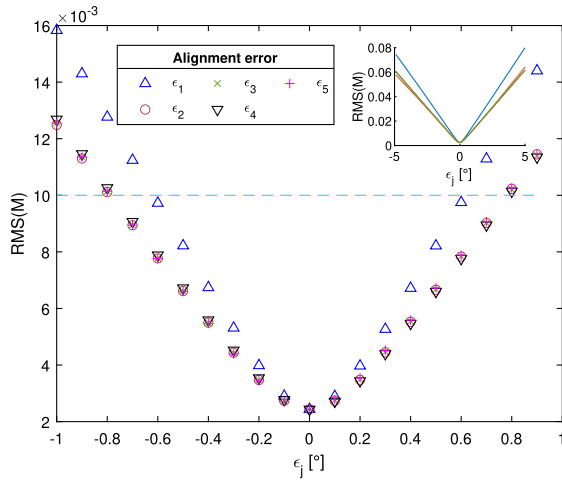


Fig. 9. Total RMS (M) error against the alignment error ϵ_j for $j = 1, 2, \dots, 5$ (first, second, third, and fourth retarder, and second polarizer, respectively) within a tolerance of $\pm 1^\circ$. The nominal orientations of the components are $\theta_1 = 45^\circ$, $\theta_2 = 0^\circ$, $\theta_3 = 0^\circ$, $\theta_4 = 45^\circ$ for the birefringent retarders, and $\theta_5 = 0^\circ$ for the second polarizer.

second polarizer is similar, for these elements immunity are achieved within a tolerance of $\pm 0.8^\circ$.

E. Gaussian Noise

A random noise distribution was added to the spectrum leaving the MMCS before entering the spectrometer. The amplitudes were in the range from 1×10^{-6} to 1, and considering a unit amplitude signal, these values correspond to a range of signal-to-noise ratio (SNR) from 1×10^6 to 1. We also included a reference without noise.

Figure 10 shows the extracted Mueller matrix, considering additive noise. It is easily observed that the MMCS is generally immune to the noise distribution added, except for the SNR < 100 (see the RMS plots in Fig. 11). However, it can be seen that the sensitivity to the noise is not the same for all the normalized Mueller matrix elements. For example, m_{13} , m_{23} , and m_{32} have a RMS error higher than the immunity condition for the SNR = 1000, whereas, m_{12} , m_{20} , and m_{30} have a total RMS

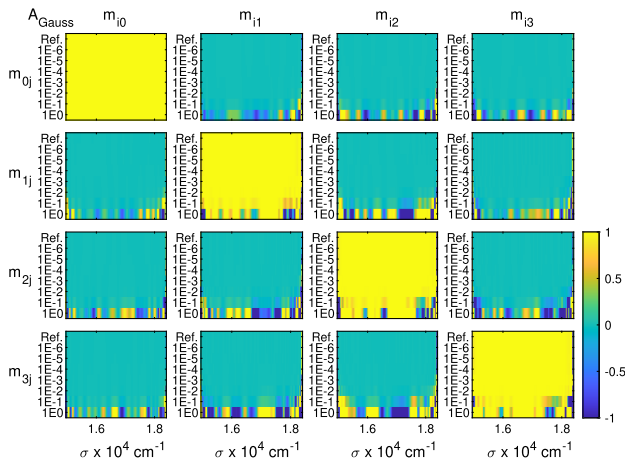


Fig. 10. Normalized Mueller matrix elements (m_{ij}/m_{00}) of the sample (air) with a configuration (1, 2, 5, 10) against the Gaussian noise distribution amplitude.

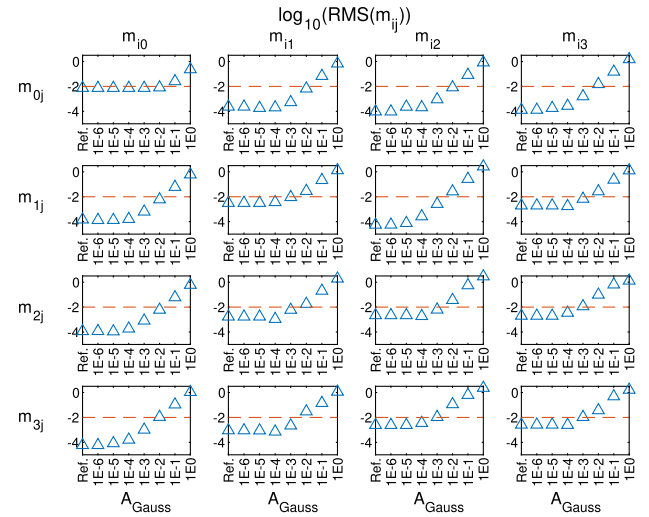


Fig. 11. RMS error of the Mueller matrix elements m_{ij} against the Gaussian noise distribution amplitude.

error that is about an order of magnitude smaller for higher SNR levels. This could be important for partial Mueller matrix polarimetry, where optimization is required for the particular Mueller matrix elements to be measured.

4. CONCLUSION

In this paper, the results of analysis and simulations of active channeled polarimeters, specifically with spectral channeling, and including some error sources and measurement noise, were presented. Two figures of merit, namely, EWW and CN, were studied as metrics of the polarimeter performance and noise immunity. The main metric to evaluate the performance of the polarimeters was the RMS.

Six configurations were compared, and three of them were known to have the minimum EWW scores [2], that is the highest immunity to additive noise. Although all six configurations had RMS scores within the immunity condition, it was observed that the configurations (1, 2, 5, 10), (2, 1, 4, 11), and (2, 1, 5, 12) offered the lowest RMS scores, between 1×10^{-4} and 1×10^{-5} for most of the unknown sample Mueller matrix elements.

The MMCS setup analyzed together with the particular data extraction algorithm used, requires a global retardance factor ≥ 4 mm in the simulation.

It is also interesting that the errors in axis alignment and retardance value in the thick retarders used in the polarimeter affect the total RMS error differently, meaning that the error tolerances are different for different retarders. Further studies are being performed to find if this behavior is general or if it depends on particular polarimeter configurations.

Overall, the extraction of the Mueller matrix elements was achieved with acceptable extraction. We are currently working on further analysis using samples with more complex polarization properties and spectral polarization variations.

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Data Availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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